RowPod v2 assembly guide

For the MakerWorld print. Board RowPod2, schematic V1.0. The printed housing, battery connector, and oar mount are the models in this listing. They are not an official Concept2 product.

1. Download the files from GitHub

1. Open **https://github.com/MHYeung/RowPod**.
2. Choose **Code**, then **Download ZIP**. Unzip the folder.
3. In **pcb/** you need:

    - **rowpod_gerber.zip** — the board. Order it as **4 layers**. Upload the zip as EasyEDA produced it. Do not merge the drill files yourself. The outline is about 30.6 mm by 33.0 mm.
    - **rowpod_v2_lcsc_bom.xlsx** — the parts order. The yellow cell is the number of boards.
    - **SCH_rowpod_SCH_2026-11-05.pdf** and **PCB_rowpod_2026-11-05.pdf** — schematic and fabrication drawing.

4. **firmware/rowpod.bin** is the program. Flash steps are in section 6.
5. The tablet app is **download/rowpod_1.5.4.apk** (version code 7, package org.rowpod.app). Install steps are **download/README.md** in that same zip.
6. The browser dashboard needs no install. Chrome or Edge: **https://mhyeung.github.io/RowPod/?v=1.5.6**. The IMU lab is lab.html?v=16.13 on that same site.

2. Order the board and the parts

Every quantity in the spreadsheet is already for one board. The yellow cell is how many boards you are building, and the order column multiplies by that. Order by the LCSC number. The manufacturer part string is copied from the EasyEDA bill of materials dated 5 November 2026.

There is no pick-and-place file in the repository. Place parts from the designators, or export a centroid file from the EasyEDA project before you send the board for assembly.

Do not order the test-point pads. On the finished board, **SCL** is the I2C clock and **TP5** is VBUS.

3. Parts that are not on the LCSC list

- **Pod housing**, one. Print it from this listing.
- **Battery connector**, one. Print it from this listing.
- **Oar mount**, one. Print it from this listing. It fits a Concept2 sleeve.
- **1-cell LiPo**, one. 3.7 V nominal. The charger is set to 400 mA, so the cell must be rated for a 400 mA charge.
- **Foam strip**, two. They sit under the cell.

4. Assemble the board

1. Fit every LCSC line. The parts are on the top side.
2. **LED1** (orange) is the charge lamp. **LED2** (blue) is the status lamp. The blue lamp has to face the window in the shell.
3. Leave **R8 at 3 kΩ**. That is the TP4056 program resistor, and it sets the charge current to 400 mA.
4. Leave **R16 and R17 at 470 kΩ**. They divide the pack voltage into GPIO2. A much smaller pair changes the reading and the boot behaviour of that pin.
5. Leave **R11 and R12 at 4.7 kΩ**. They are the I2C pull-ups. The IMU address is 0x6A.
6. Fit **L2** (3 nH) and both **0 Ω** links **R13 and R14**. They are the antenna match for the CA-C03.

7. There is a 40 MHz crystal and no 32 kHz crystal.

5. Cell, shell, and mount

1. Solder the cell to **H1**. The silkscreen is **B+** and **B–**. B+ is the cell positive. Check the silkscreen before you tin the pads. A reversed cell is connected straight to the charger.
2. Put the two foam strips under the cell so the pack cannot press on the parts.
3. Close the printed housing so the USB-C plug, the blue LED, and the user button **SW1** line up with the openings.
4. Clip the housing into the printed oar mount.

The three buttons on the board are:

- **SW1** — user button, GPIO0. Hold it about 2 seconds to wake the pod. A short press while the pod is asleep does nothing.
- **RST** — resets the chip.
- **B** — boot, GPIO9. Used only when flashing.

6. Flash the firmware

A new ESP32-C3FH4 does not contain RowPod. Use **firmware/rowpod.bin** from the zip in section 1. That one file is the bootloader, the partition table, and the app.

On Windows, install Espressif Flash Download Tool. Choose chip type **ESP32-C3**. Add **rowpod.bin** at address **0x0**. Set SPI mode **DIO**, SPI speed **80 MHz**, and tick **DoNotChgBin**. Plug in USB-C, select that COM port, hold **B**, tap **RST**, release **B**, then click **START**.

Compiling the image yourself needs ESP-IDF 6.0.1. Those commands are in the repository README, after the flash steps.

7. First wake

1. Install **rowpod_1.5.4.apk** on the Android tablet, or open the dashboard link from section 1 on a computer running Chrome or Edge. There is no iPhone build.
2. Hold **SW1** for about 2 seconds. The blue LED breathes, and the pod advertises as **C3-T-** plus four characters.
3. In the app or the dashboard, choose **Add pod** and select that name. The LED stays dim while connected.
4. With nothing connected, the pod sleeps after about 3 minutes and the LED goes out. Hold SW1 again to wake it.

Charge current, button names, and the LCSC numbers come from schematic ROWPOD_SCH V1.0 and the fabrication PDF exported 5 November 2026.